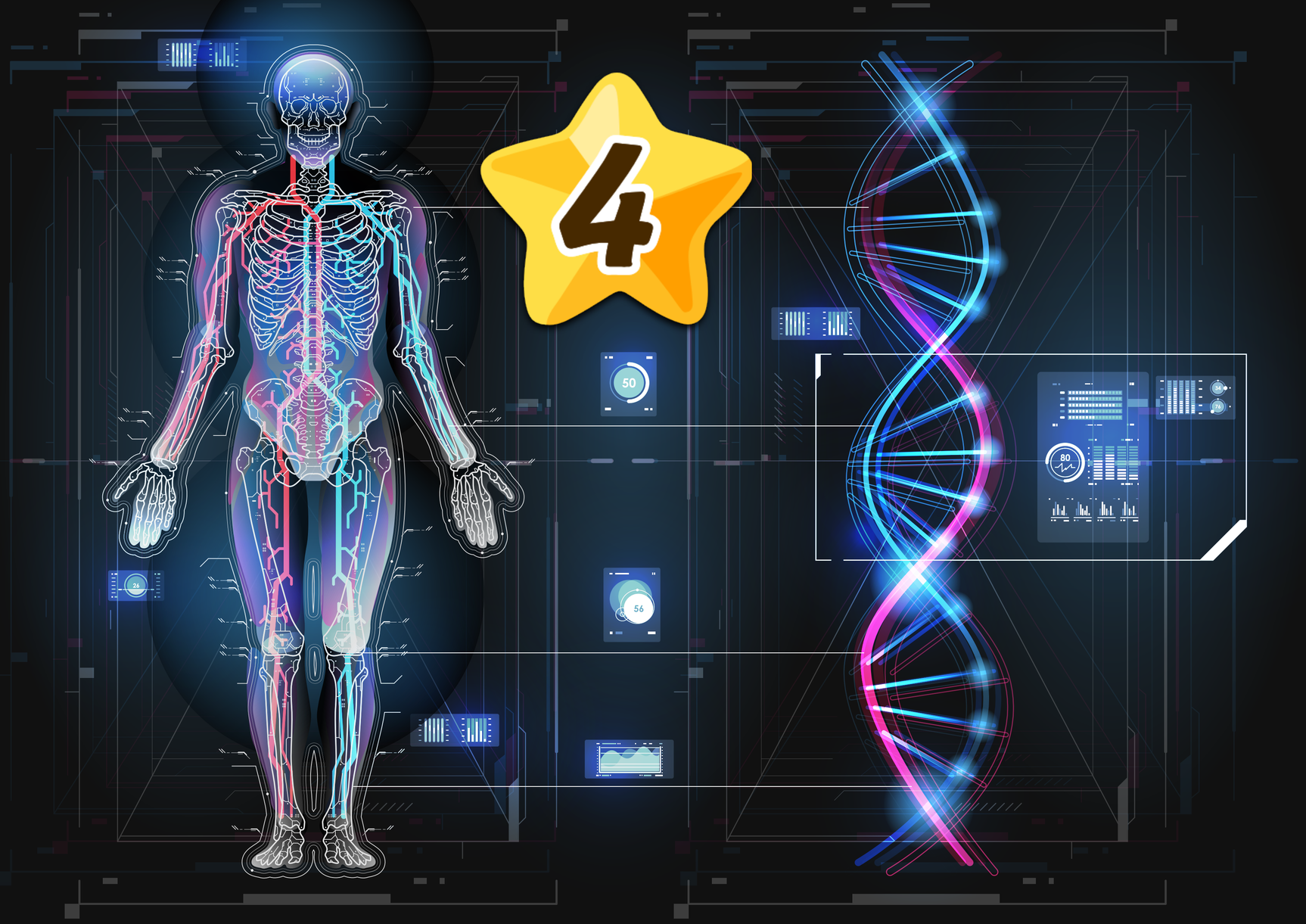


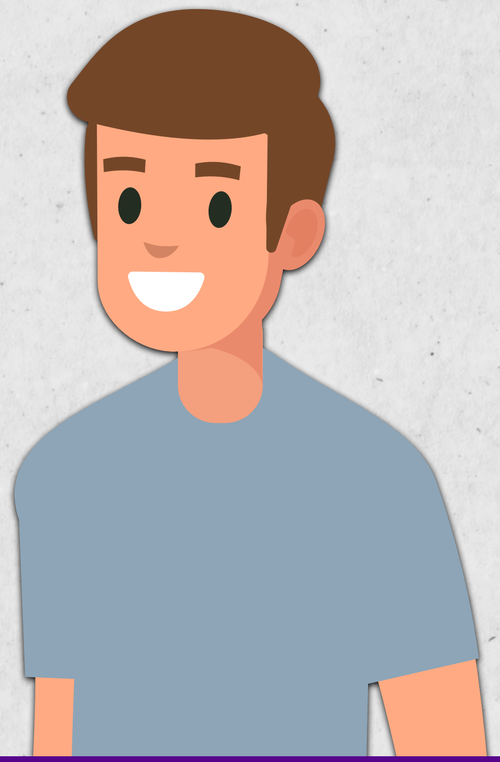
HUMAN DIGESTION, ABSORPTION, AND TRANSPORT

LEARNING OBJECTIVES

01 Review the organs and functions of the digestive system

02 Explore nutrient absorption

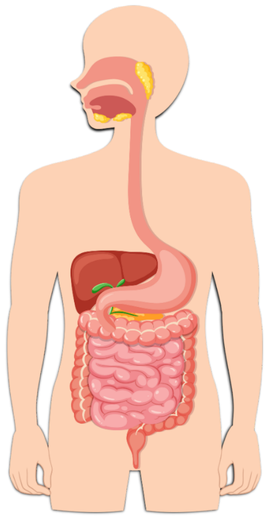
03 Explore elimination

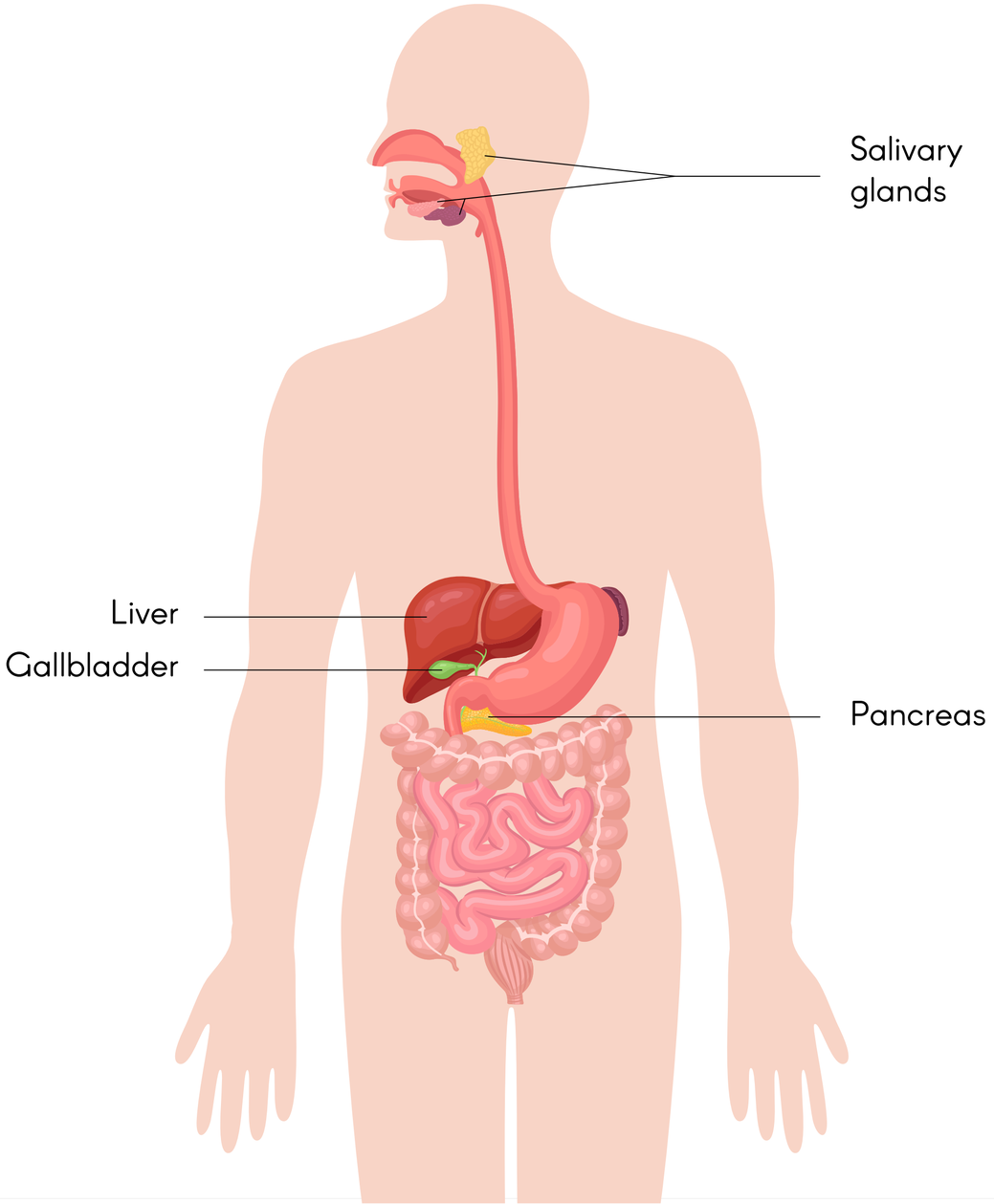


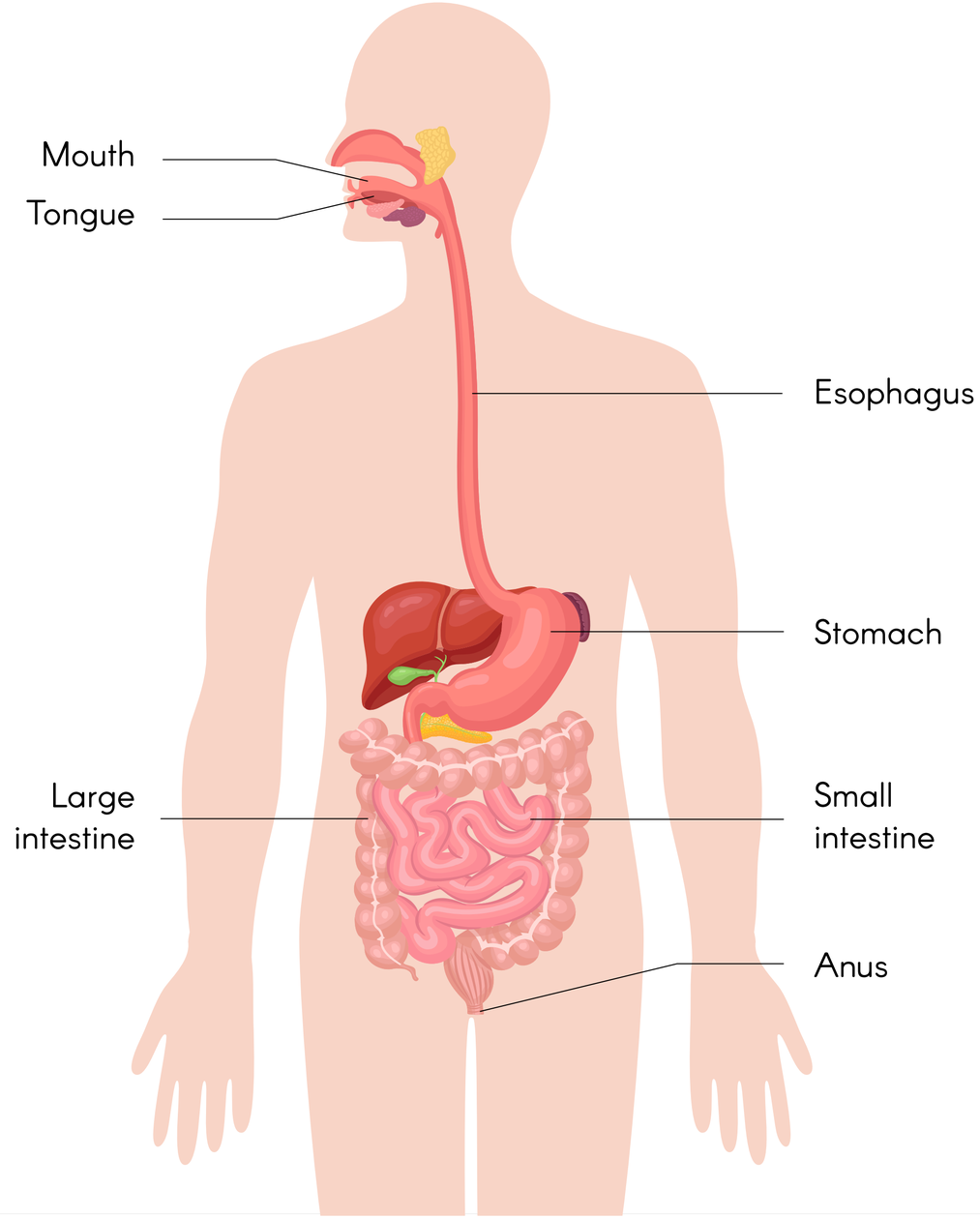
THE DIGESTIVE SYSTEM

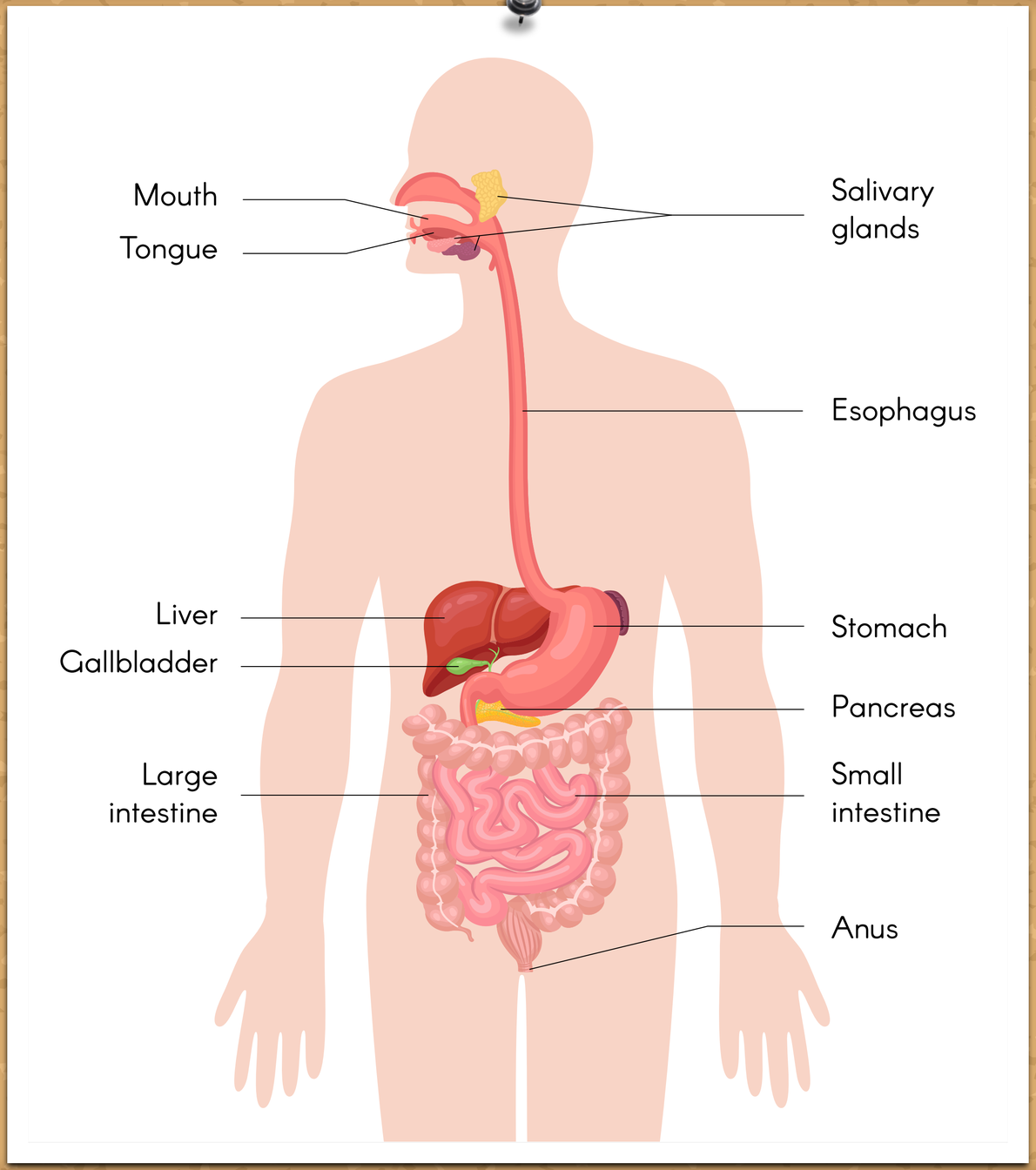
- **Responsible for breaking down food and absorbing nutrients**
  - ➡ Also responsible for the elimination of waste products
- **Digestion is the process of breaking down food molecules into nutrients**
- **Absorption is the process of nutrient uptake from the gastrointestinal tract**

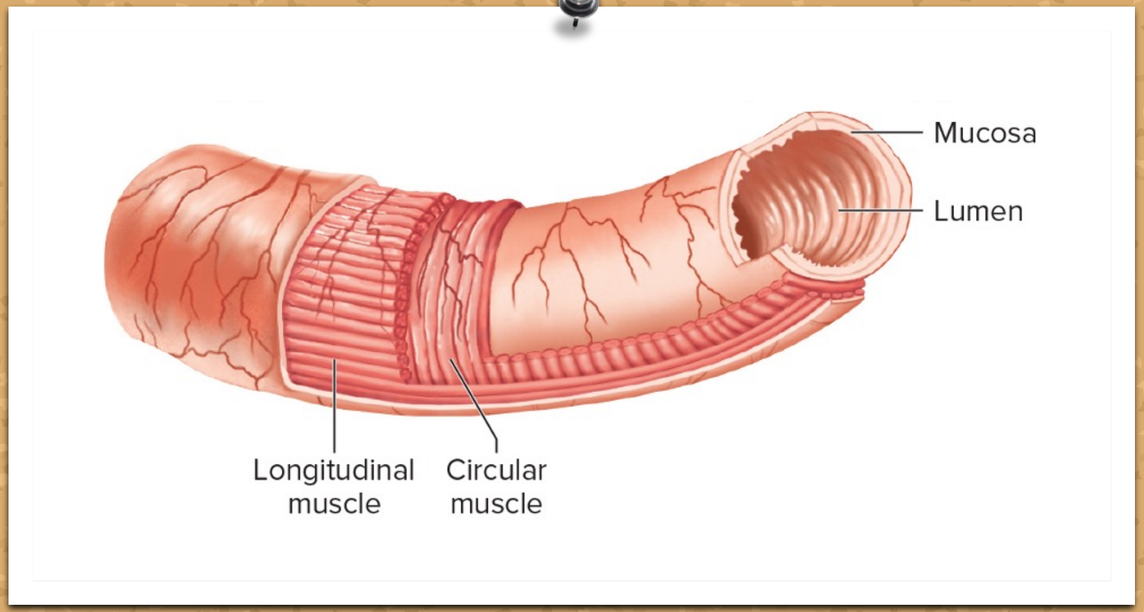
A hollow, muscular tube that extends from mouth to anus

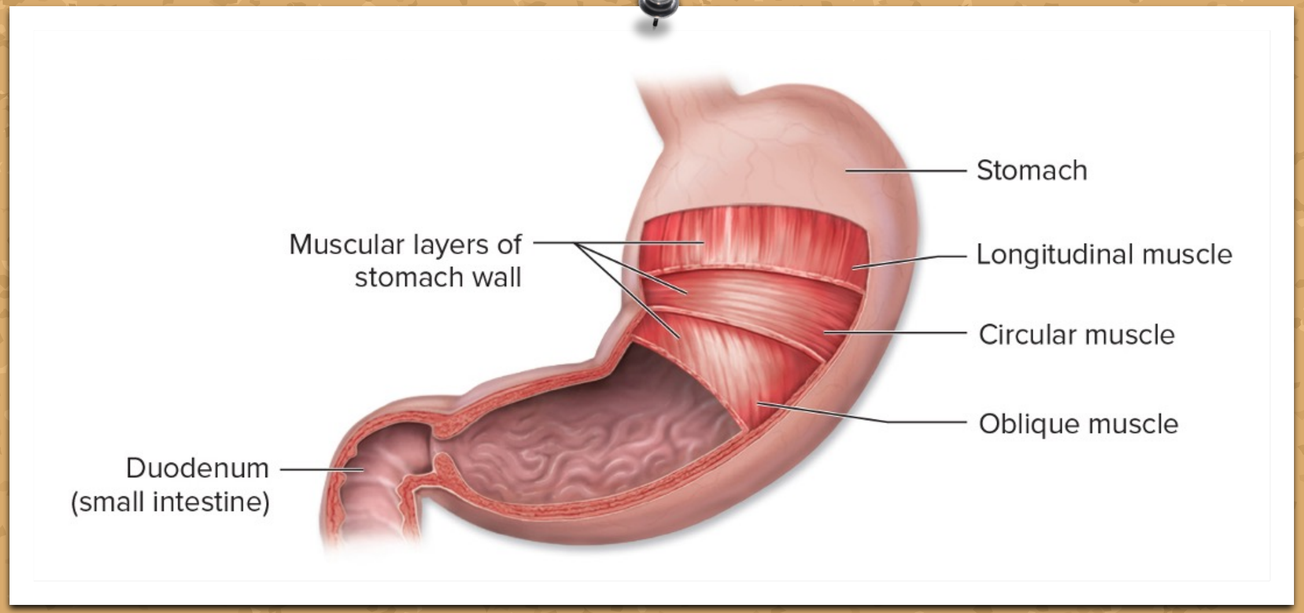


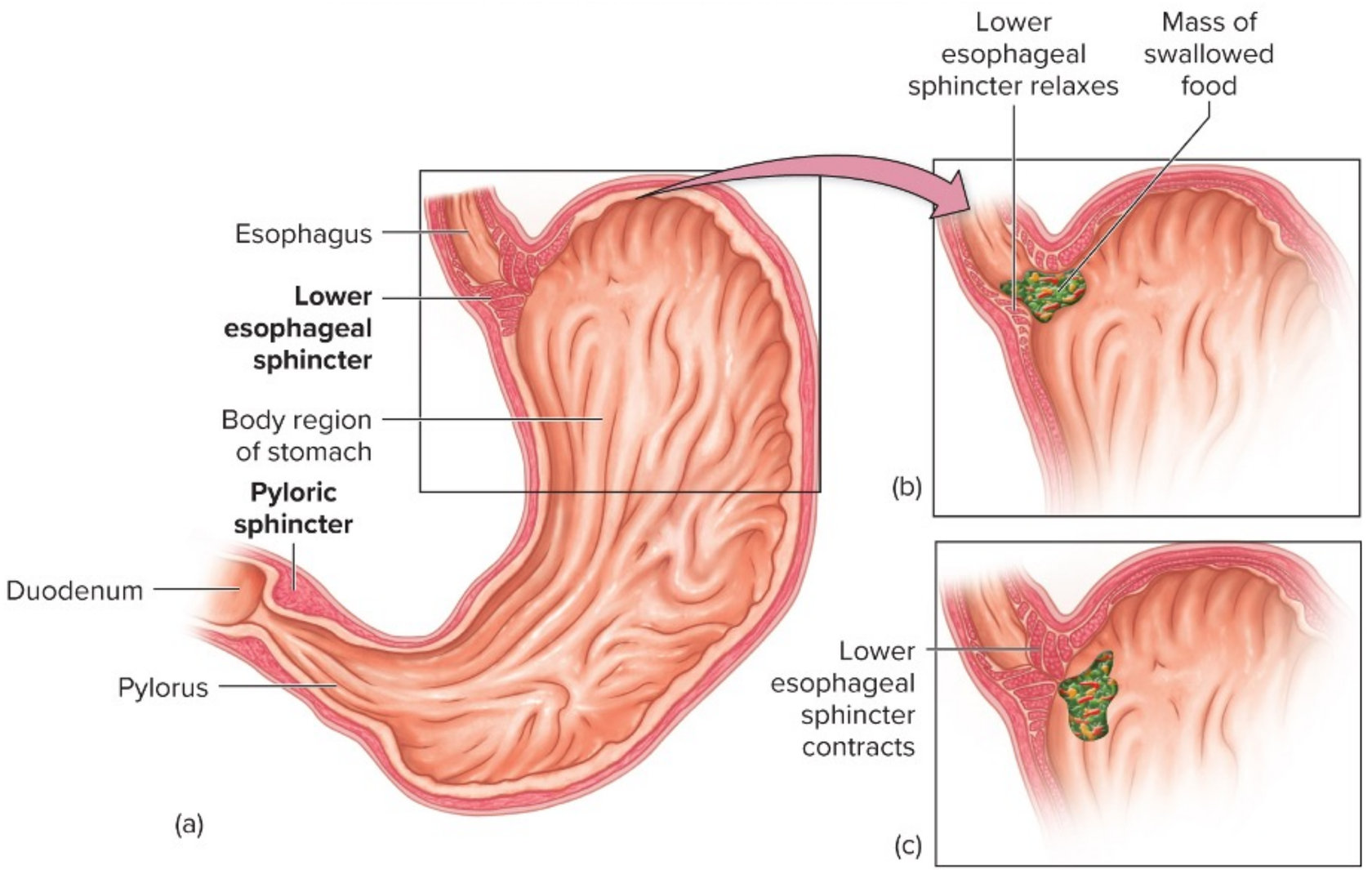

Salivary glands
Liver
Gallbladder
Pancreas


Mouth
Tongue
Esophagus
Stomach
Large intestine
Small intestine
Anus


Mouth
Tongue
Salivary glands
Esophagus
Liver
Gallbladder
Stomach
Pancreas
Large intestine
Small intestine
Anus


Mucosa
Lumen
Longitudinal muscle
Circular muscle


Stomach
Muscular layers of stomach wall
Longitudinal muscle
Circular muscle
Oblique muscle
Duodenum (small intestine)

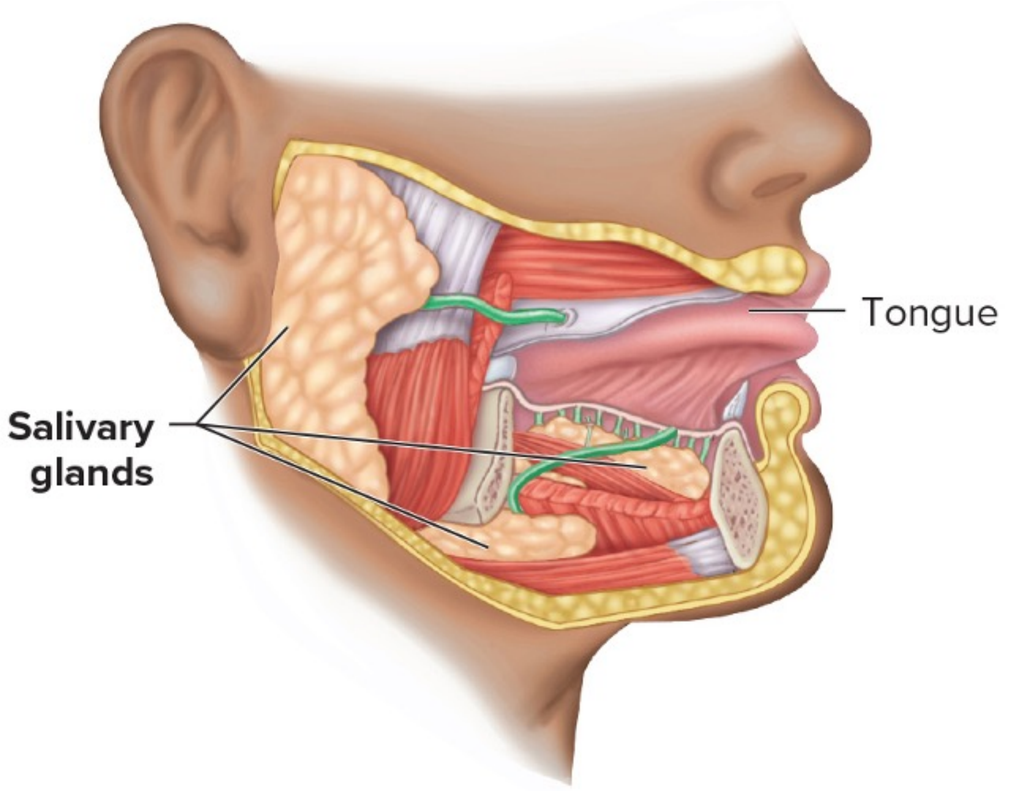

Lower esophageal sphincter relaxes
Mass of swallowed food
Esophagus
Lower esophageal sphincter
Body region of stomach
Pyloric sphincter
Duodenum
Pylorus
(a)
(b)
Lower esophageal sphincter contracts
(c)

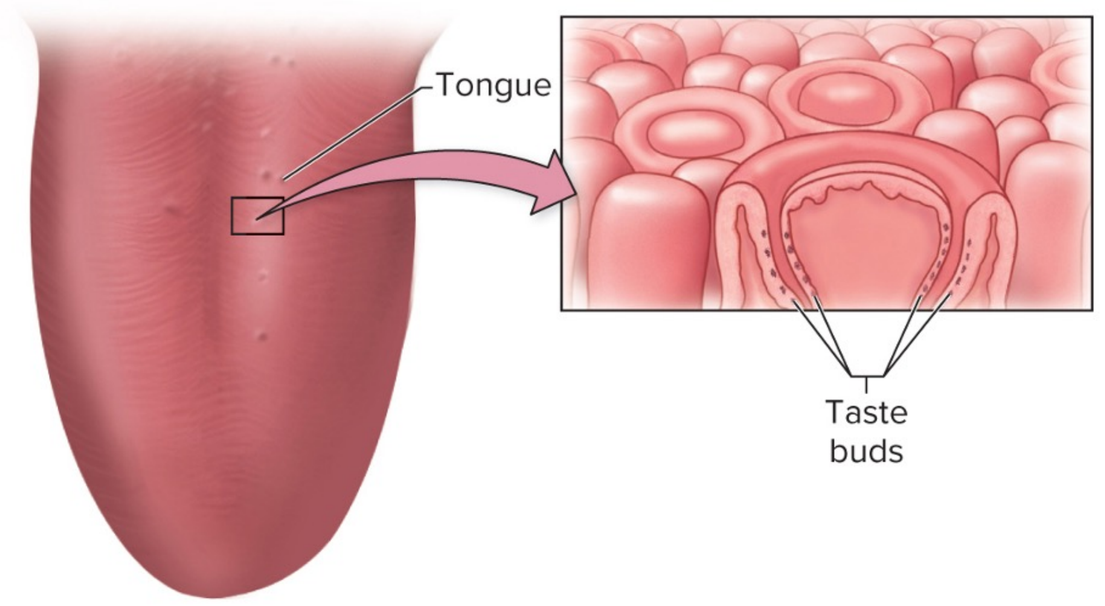
MECHANICAL VS. CHEMICAL DIGESTION

- **Mechanical digestion is the physical act of breaking down food**
  - ➡ Example: chewing a cracker into smaller pieces
- **Chemical digestion relies on the activity of digestive juices**
  - ➡ Includes enzymes, hydrochloric acid, and bile
  - ➡ Produced by the accessory organs, stomach, and small intestine

MOUTH

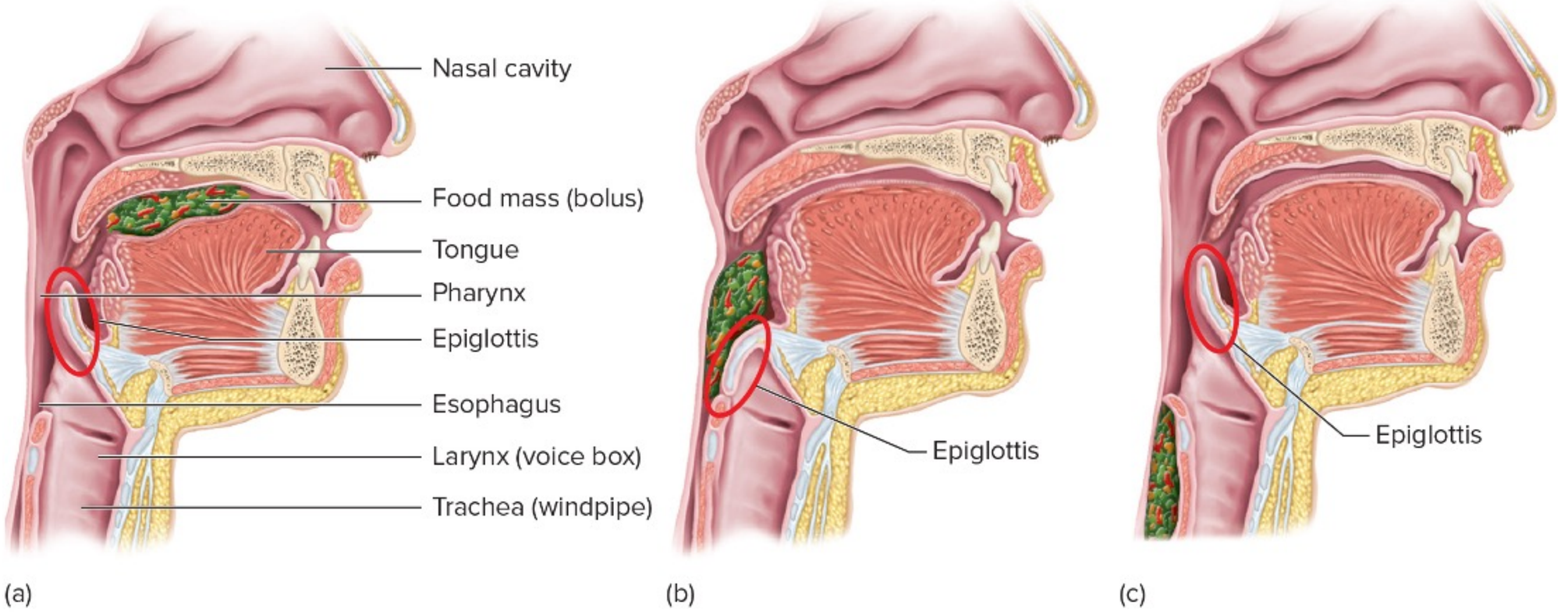
- **Where mechanical and chemical digestion begin**
  - ➡ Mastication makes foods easier to swallow and increases its surface area
  - ➡ Several salivary glands secrete saliva into the oral cavity
  - ➡ Saliva contains water, mucus, and a few digestive enzymes
    - ⋆ *Example: Salivary amylase begins the breakdown of starch*
- **Plays a role in sensing the taste, temperature and texture of foods**
  - ➡ Taste buds are specialized cells found on the top of the tongue

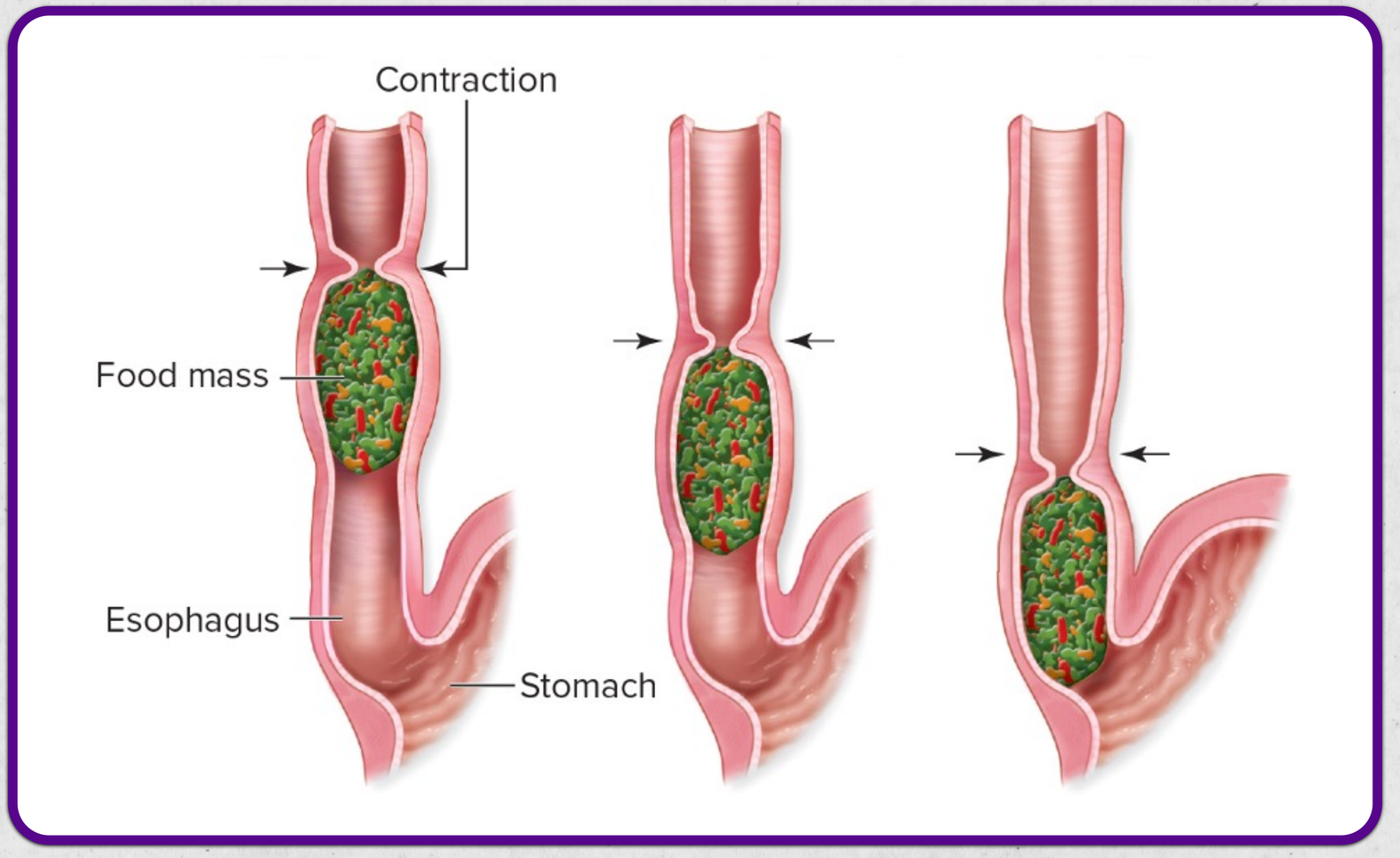

Tongue
Salivary glands


Tongue
Taste buds

ESOPHAGUS

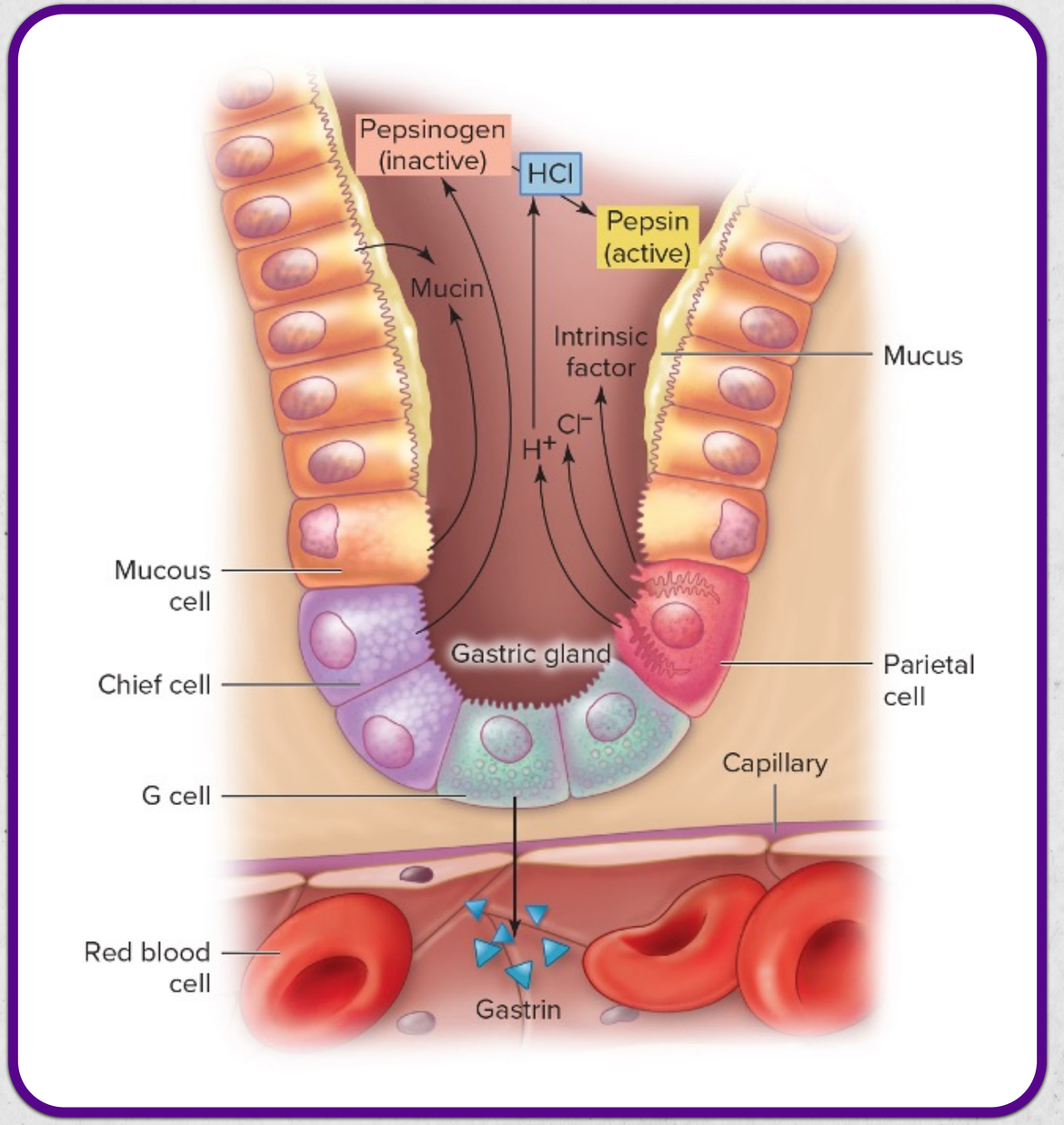
- **Organ that transfers a bolus from the mouth to the stomach**
  - ➡ Requires a coordinated effort to protect the larynx and trachea
- **Swallowing stimulates peristalsis**
  - ➡ Helps move materials through the remainder of the gastrointestinal tract
- **The lower esophageal sphincter prevents the backflow of stomach contents**
  - ➡ Acid reflux occurs when it fails to close properly


Nasal cavity
Food mass (bolus)
Tongue
Pharynx
Epiglottis
Esophagus
Larynx (voice box)
Trachea (windpipe)
(a)
Epiglottis
(b)
Epiglottis
(c)


Contraction
Food mass
Esophagus
Stomach

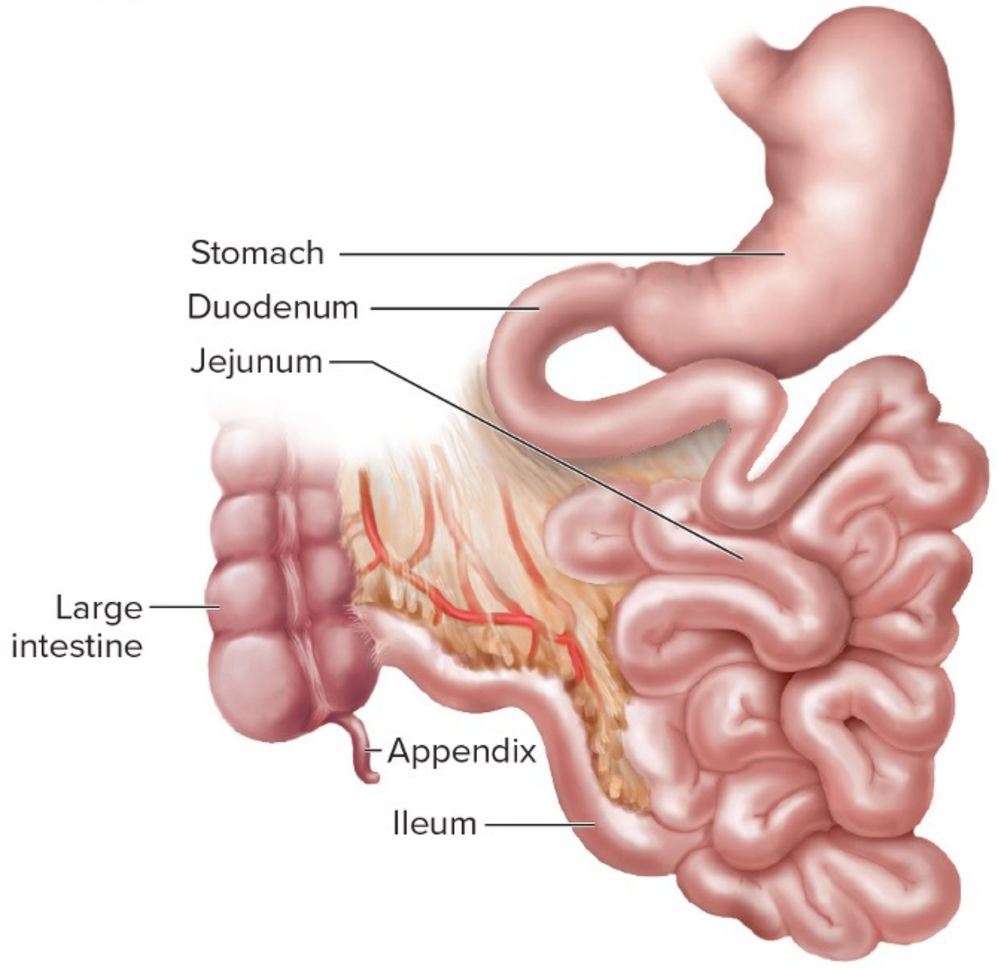
STOMACH

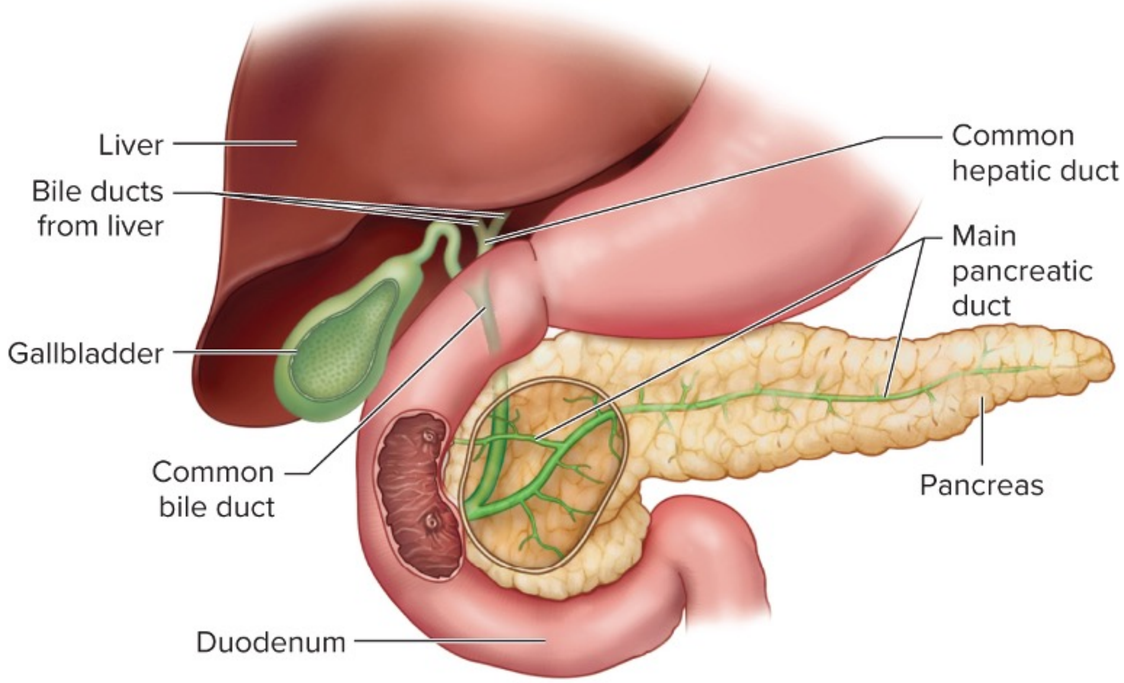
- **A muscular sac that stores and mixes food through churning movements**
  - ➡ Contains glands that secretes gastric juice
  - ➡ Gastric juice mixes with food to form chyme
- **The pyloric sphincter regulates the flow of chyme into the small intestine**
  - ➡ Emptying time depends on the contents and size of the meal
    - * *Example: Fiber slows gastric emptying*


Pepsinogen (inactive)
HCl
Pepsin (active)
Mucin
Intrinsic factor
Mucus
Cl⁻
H⁺
Mucous cell
Gastric gland
Chief cell
Parietal cell
Capillary
G cell
Red blood cell
Gastrin

SMALL INTESTINE

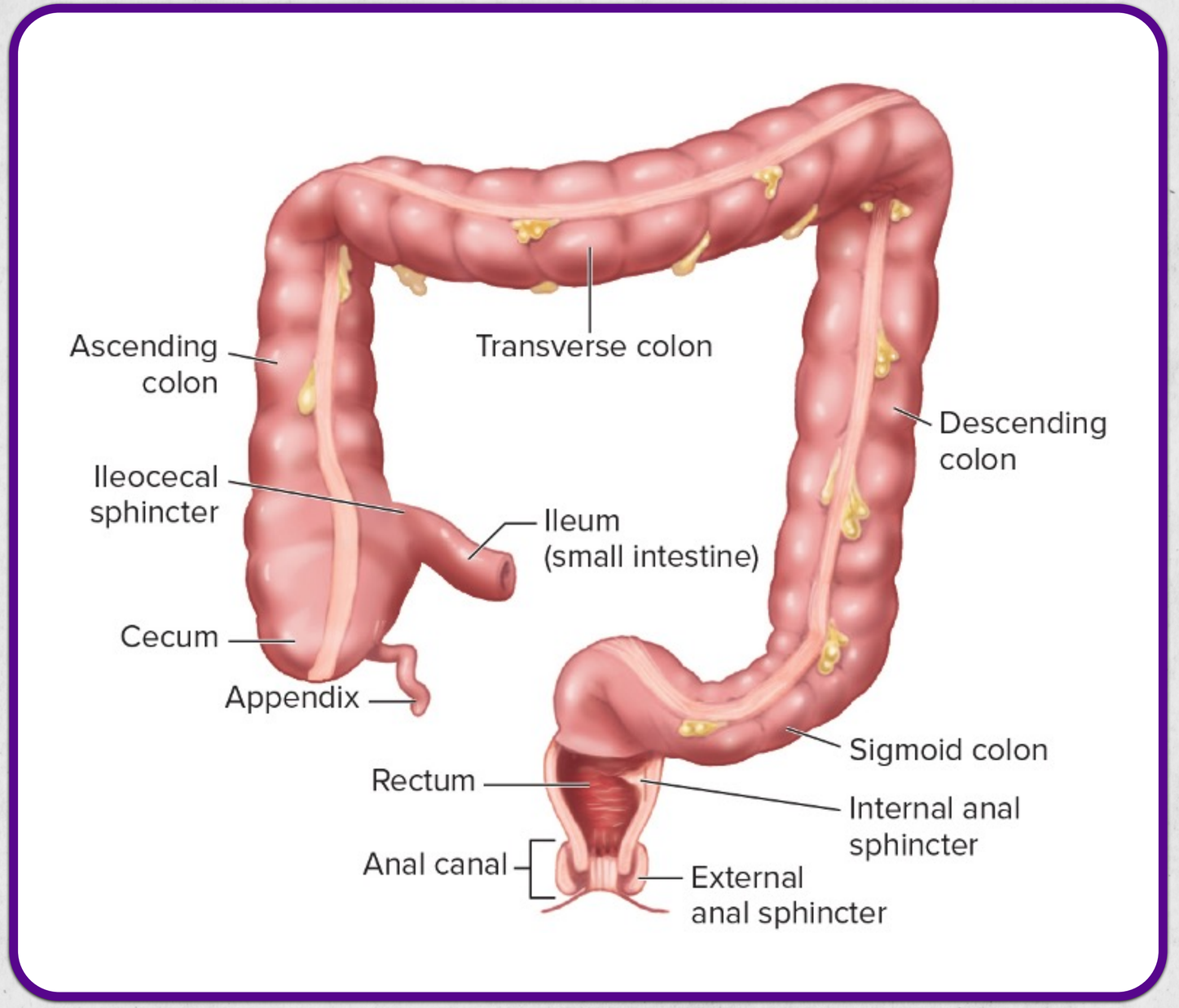
- **Coiled, hollow tubes that extends from the stomach to the large intestine**
  - ➡ Longest component of the gastrointestinal tract (~20 feet)
  - ➡ Called “small” because of the tube’s diameter (~1 inch)
- **Primary site for nutrient digestion and absorption**
  - ➡ Digestion relies on secretions from the liver, gallbladder, and pancreas
    - ⋆ *Example: Bile is needed for fat digestion and absorption*
- **The ileocecal sphincter controls the flow of contents into the large intestine**
  - ➡ Also prevents the backflow of contents into the small intestine


Stomach
Duodenum
Jejunum
Large intestine
Appendix
Ileum

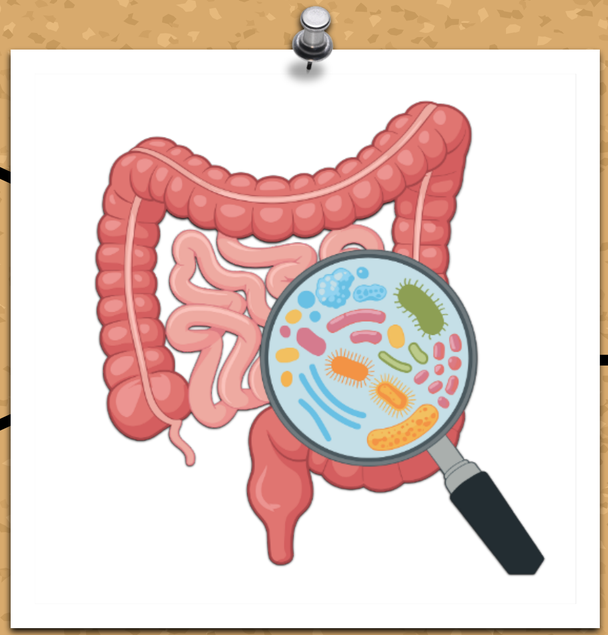

Liver
Bile ducts from liver
Gallbladder
Common bile duct
Duodenum
Common hepatic duct
Main pancreatic duct
Pancreas

LARGE INTESTINE

- **Stores and concentrates undigested material and prepares it for excretion**
  - ➡ Much shorter than the small intestine (~5 feet) but has a larger diameter
  - ➡ Also responsible for the absorption of water and some minerals
- **Home to large populations of microorganisms called the gut microbiota**
  - ➡ Feed on undigested food substances, such as fiber
  - ➡ Produce neurotransmitters and vitamins (vitamin K, vitamin B-12, and thiamin)
  - ➡ Help to develop and regulate the immune system
- **Dysbiosis refers to an undesirable imbalance of microbial populations**
  - ➡ Can result from poor diet, antibiotic use, and exposure to environmental toxins


Ascending colon
Transverse colon
Descending colon
Ileocecal sphincter
Ileum (small intestine)
Cecum
Appendix
Sigmoid colon
Rectum
Internal anal sphincter
Anal canal
External anal sphincter

Probiotics



Prebiotics

Primary Organ	Function
Mouth	*Chews and mixes food with saliva*
Esophagus	*Passes food into the stomach*
Stomach	*Adds acid, enzymes, and fluid; Churns, mixes, and grinds food to a liquid mass*
Small Intestine	*Primary site of digestion and absorption*
Large Intestine	*Removes water and minerals; Prepares waste for removal from the body*

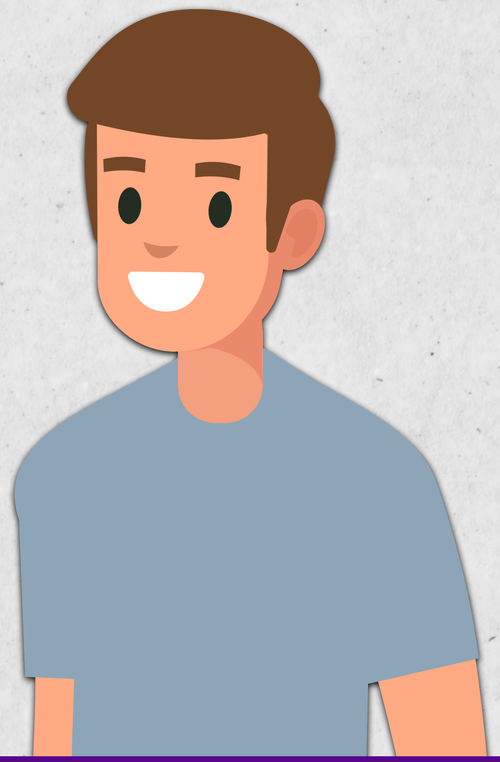
Accessory Organ	Function
Salivary Glands	*Produces digestive enzymes; Saliva moistens the food*
Liver	*Manufactures bile, a substance that facilitates fat digestion*
Gallbladder	*Stores bile until it is needed for fat digestion*
Pancreas	*Produces digestive enzymes; Helps to neutralize stomach acid*

LEARNING OBJECTIVES

01 Review the organs and functions of the digestive system

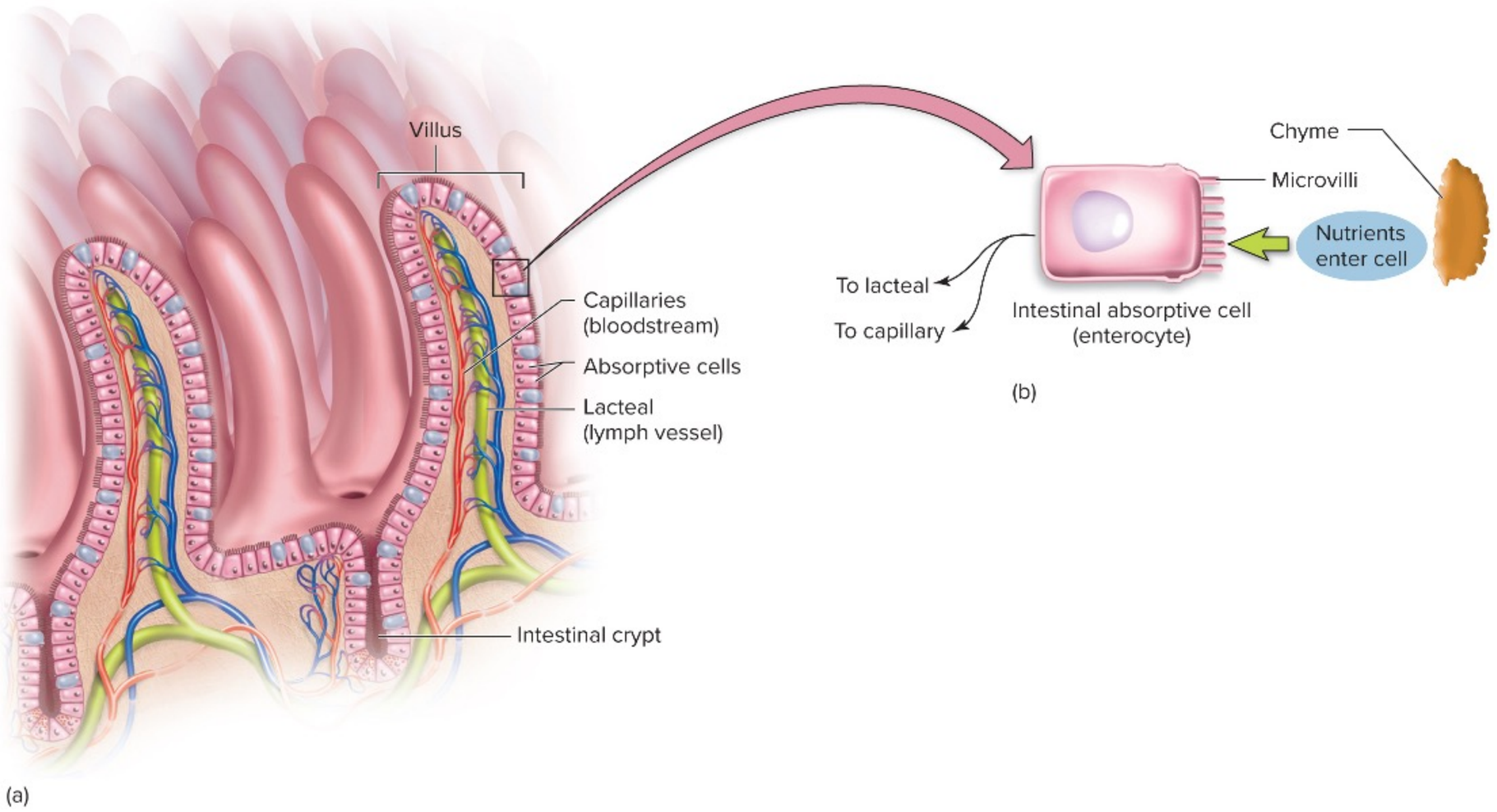
02 Explore nutrient absorption

03 Explore elimination



ABSORPTION

- **Occurs when nutrients leave the digestive tract and enter the cells lining it**
  - ➡ The primary site of absorption is the small intestine
  - ➡ It is an efficient process made possible by the presence of villi and microvilli
  - ➡ Cells that line the outer layer of each villus are called enterocytes

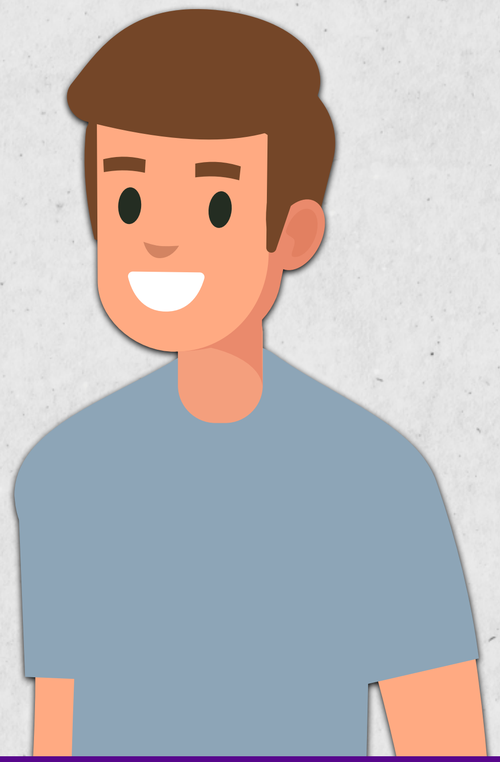

Villus
Capillaries (bloodstream)
Absorptive cells
Lacteal (lymph vessel)
Intestinal crypt
(a)
Chyme
Microvilli
Nutrients enter cell
To lacteal
To capillary
Intestinal absorptive cell (enterocyte)
(b)

ABSORPTION (CONTINUED)

- **Nutrients are absorbed directly into the blood or lymph**
  - ➡ **Water-soluble nutrients enter the blood**
    - * *Carbohydrates*
    - * *Protein*
    - * *Minerals*
    - * *Vitamin C and the B-vitamins*
  - ➡ **Fat-soluble nutrients enter the lymph**
    - * *Fat*
    - * *Vitamins A, D, E, and K*
- **Water-soluble nutrients travel to the liver via the <u>hepatic</u> <u>portal</u> <u>vein</u>**
- **Lymph empties into the subclavian vein near the heart**

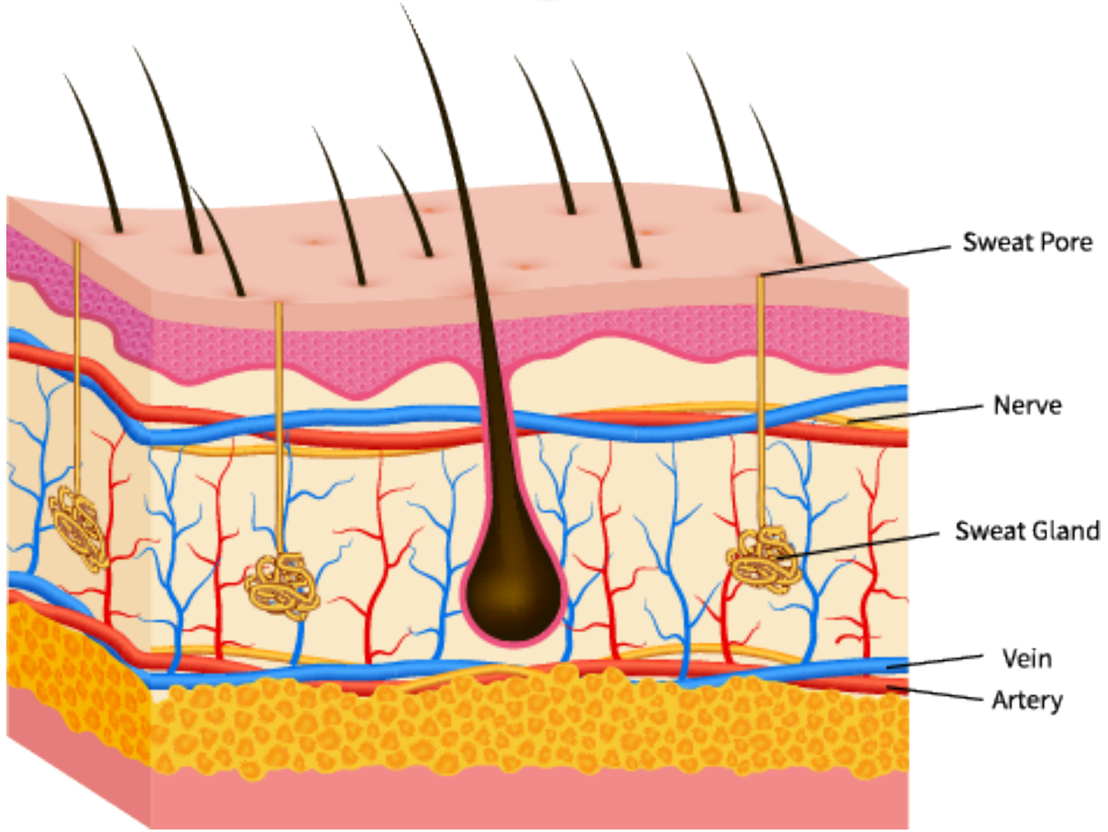
LEARNING OBJECTIVES

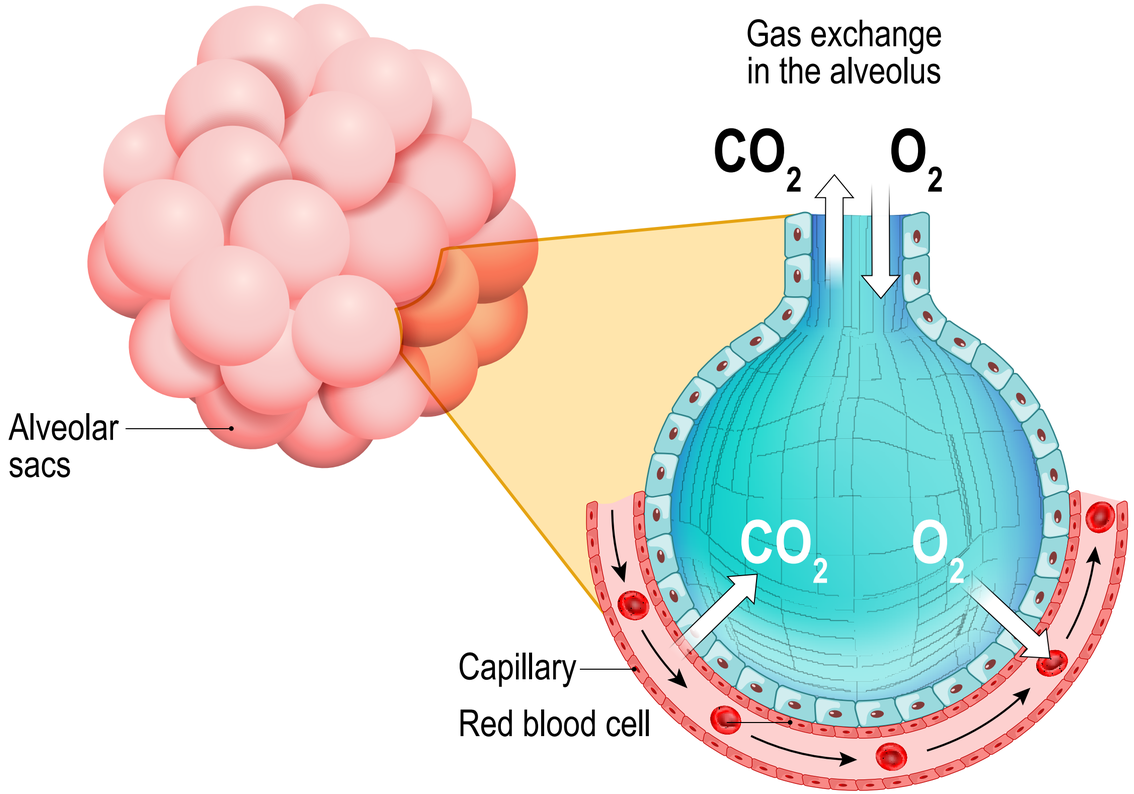
- [x] 01 Review the organs and functions of the digestive system
- [x] 02 Explore nutrient absorption
- [ ] 03 Explore elimination

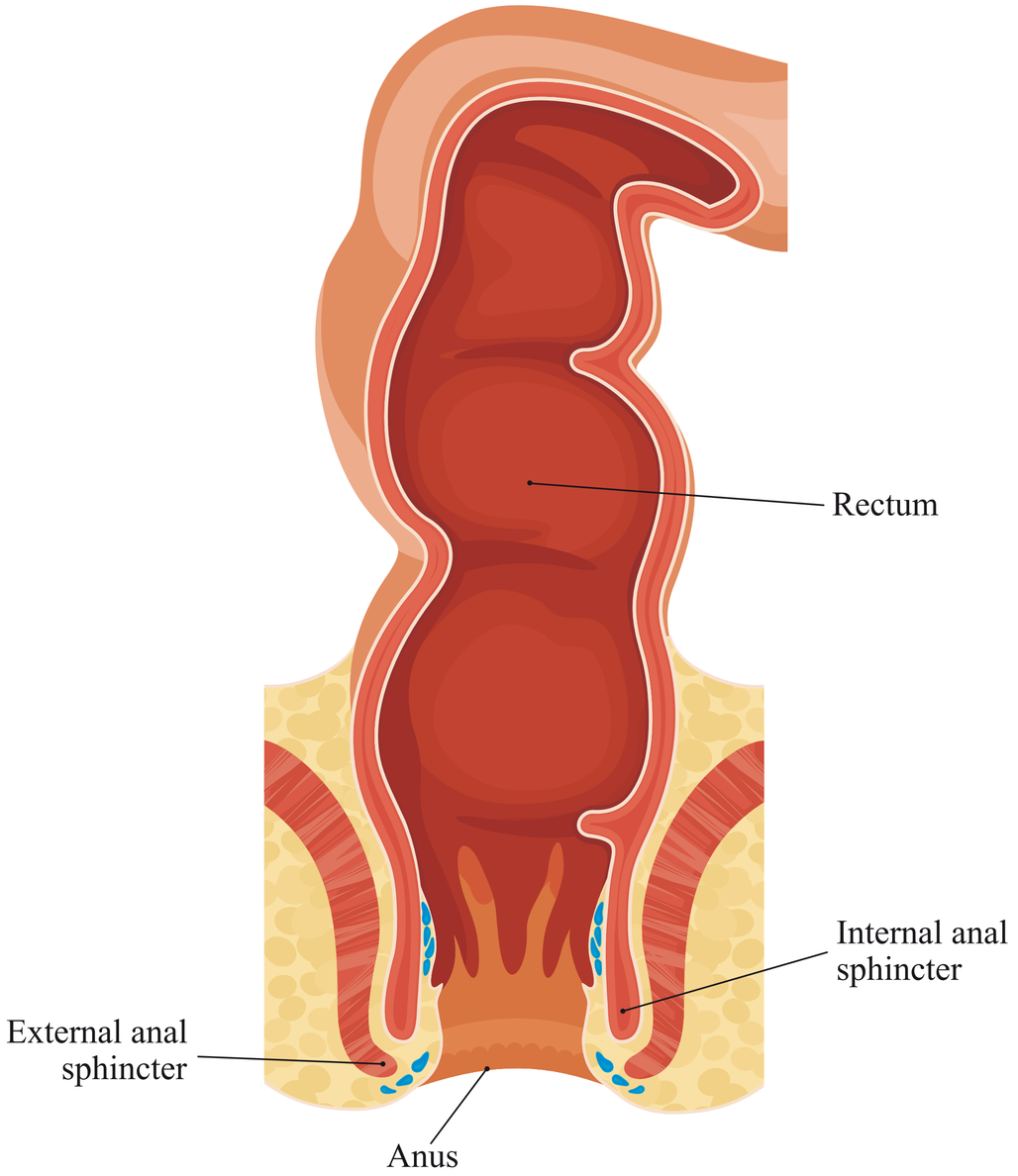


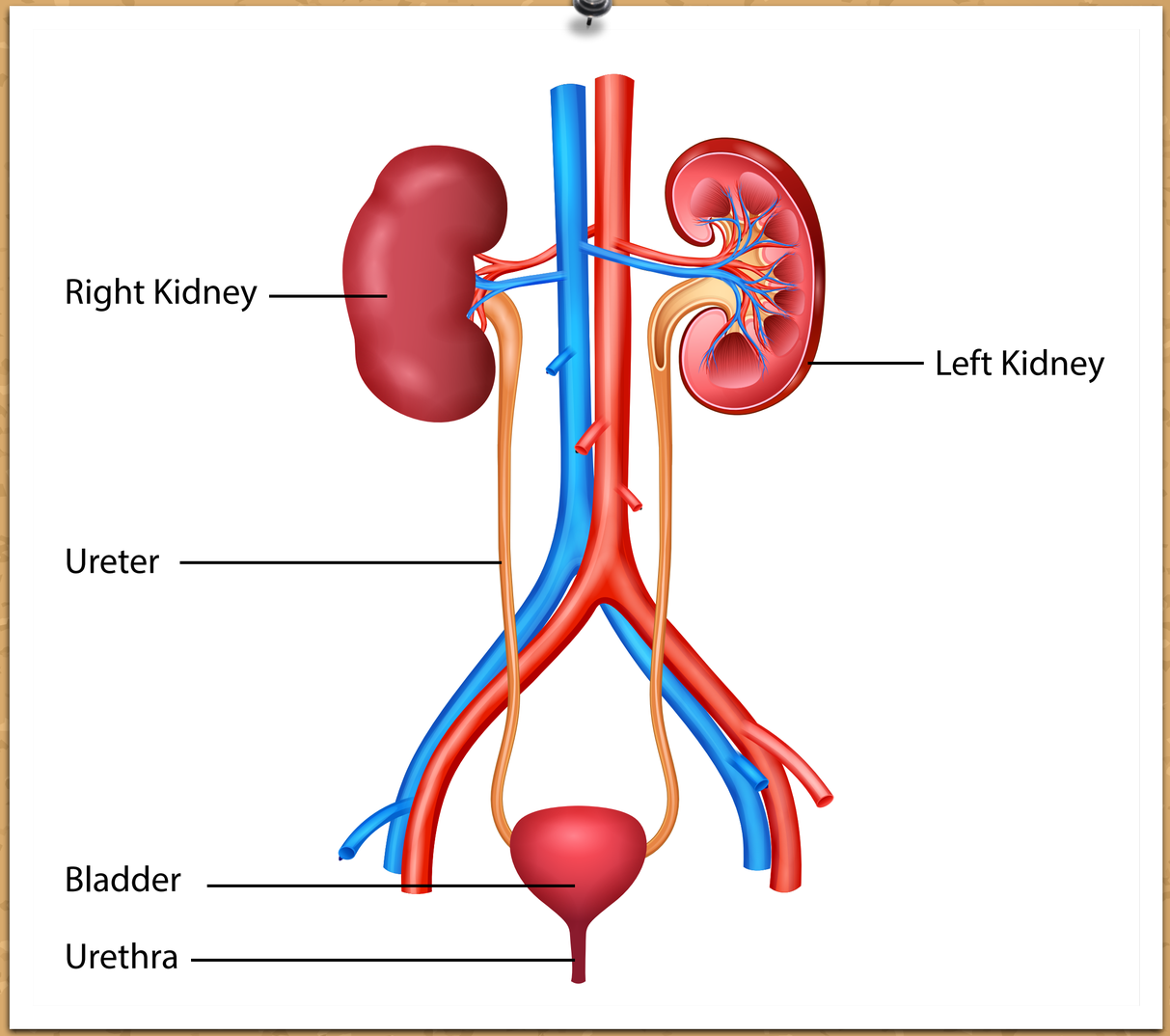
ELIMINATION

- **Waste formation is a normal byproduct of metabolic processes**
  - ➡ Organs of elimination include the skin, lungs, liver, large intestine, and kidneys
- **Urination and defecation are major waste removal processes**
  - ➡ Kidneys filter the blood and produce urine
  - ➡ Liver filters the blood and packages waste products in bile
  - ➡ The rectum stores feces and releases it through the external anal sphincter


Sweat Pore
Nerve
Sweat Gland
Vein
Artery

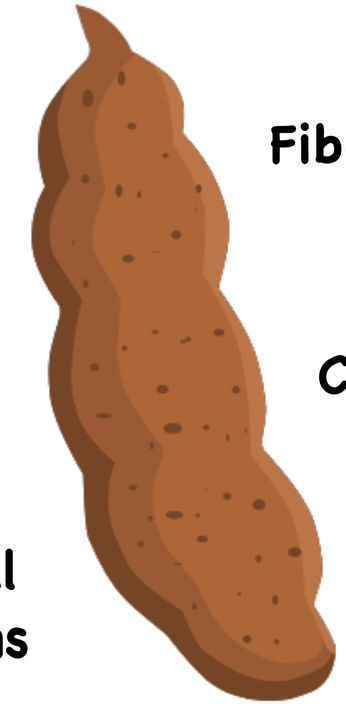

Gas exchange
in the alveolus
CO_2
O_2
Alveolar
sacs
CO_2
O_2
Capillary
Red blood cell


Rectum
Internal anal sphincter
External anal sphincter
Anus


Right Kidney
Left Kidney
Ureter
Bladder
Urethra

Water

Fiber

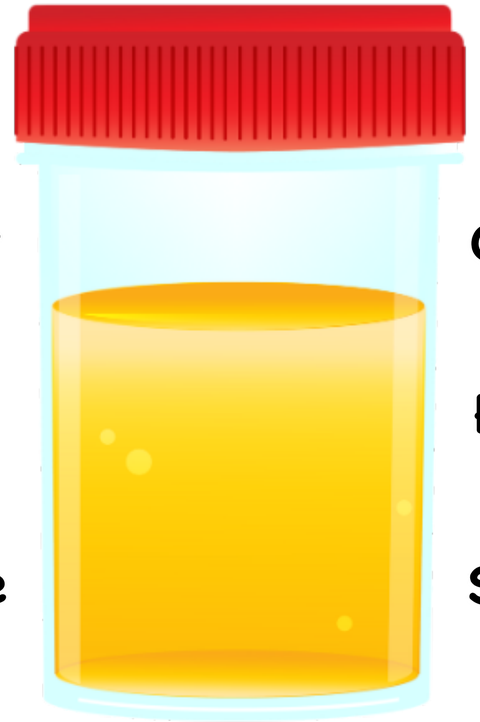


Bacteria

Cell Debris

Intestinal Secretions

Bile



Water

Chloride

Urea

Potassium

Creatinine

Sodium

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